

Numbers about the logs

counts, error rates, lag – built and running

The log text itself

the words in the line – nothing reads them

02 Three ways to read the text

None is chosen. All three are open.

TEMPLATING ALONE

Collapse a line to its skeleton

```
Processing timeslice:41
Processing timeslice:<N>
```

Millions of lines become
a few thousand templates.
A template never seen
before is a free signal.

TRADE-OFF

The signal runs out. Software prints
only the messages in its source. Once
every template is seen, it stops firing.

k-NN ALONE

Nearest-neighbour distance on lines

```
k-NN: k nearest neighbours.
Distance is the score.
```

Turn each line into a vector.
Score a new line by how far
it is from the nearest line
already seen.

TRADE-OFF

Rejected on cost. Every line runs
through a model at ingest. Millions of
vectors sit in memory already rationed.

TEMPLATES, THEN k-NN

The two ideas, fused

```
Templates, then distance.
A few thousand vectors.
```

Extract the skeletons first.
Then run the distance test
on the templates, and
not on the lines.

TRADE-OFF

It fixes the cost. It inherits some of
the saturation. Our own document
names this one as still open.

03 The proposed pipeline

On the worker

Extract the template

no model, string work
cheap on the worker

Count each template

one counter per template
one small table

push

In the storage tier

The template vocabulary

every template seen so far
held once, not per worker

The scoring

one of the three ways above
on the storage tier only

Some work here, but not much

The worker pushes its template counts upward,
and any template it has never seen before.
Nothing polls the worker. No model runs on it.